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EEE4440 — Power Electronics in Energy Systems

Literature Review on High-Gain SEPIC Converters

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19 May 2026

1. Introduction

The single-ended primary-inductor converter (SEPIC) is a fourth-order non-isolated DC-DC topology with simultaneous step-up and step-down capability and continuous input current. Those two properties together make it a natural interface for PV panels, fuel cells and battery banks. Its classical voltage gain of $D/(1-D)$ is the limitation: for low-voltage renewable sources in the 12–60 V range feeding 200–400 V DC microgrids, or for high-voltage industrial loads such as pulsed electric field water treatment, the classical formula simply does not reach.

Recent literature extends the basic SEPIC in several directions. Switched inductor and capacitor cells. Voltage lift cells. Coupled inductors. Quadratic stages. Interleaved and multiport structures. Each direction adds gain at a different cost in components, control complexity, or magnetic design effort. This report covers eight peer-reviewed articles published between 2023 and 2025. The summaries are written in a deliberately varied style, with each contributor approaching their two assigned papers from a different reading angle rather than following a uniform template. A comparison table follows the summaries, and the discussion section interprets the patterns. The discussion also introduces a derived per-device gain metric not present in the source articles, which proves useful for grouping the topologies along a single axis.

2. Individual Article Summaries

2.1 Baran Şakir Athğ (Student ID: 2003998)

Article 1 — Baladhandapani & Arul Prasanna (2025): Interleaved SEPIC-Cuk for PMSM-Based EVs

A photovoltaic-fed PMSM drive for electric vehicles sets the operational backdrop for this study. The conventional single-stage DC-DC interface is replaced by a hybrid SEPIC-Cuk converter, and the tracking layer is rebuilt around a Satin Bowerbird Optimization–Support Vector Machine algorithm. The power stage is interleaved: two phases driven 180° out of phase, each combining SEPIC continuous input current with Cuk continuous output current. The device list contains two MOSFETs, four inductors, three capacitors, and two diodes.

Two design choices stand out. The first is the dual-side current continuity, which addresses both PV-side ripple (a perennial concern for module efficiency) and motor-side current quality. The second is the substitution of conventional P&O tracking with a population-based search, an unusual choice for a converter-centric paper. The authors report converter efficiency of 92.02 % under simulation, an MPPT tracking efficiency of 97.82 %, and prototype waveforms showing 65 V boosted to 350 V driving the PMSM at 1500 rpm.

The voltage gain derivation gives $V_o/V_{PV} = 2/(1-D)$, and the comparative table cites a measured gain of 10.2 at $D = 0.585$. Substituting that duty into the formula returns about 4.82, not 10.2, and the paper does not reconcile the two figures. This kind of internal inconsistency is uncomfortable in a journal publication. It forces the reader to choose between trusting the analytical derivation and trusting the experimental table, with no clear guidance from the authors. The MPPT side is the stronger half of the article. The converter analysis would benefit from a second editorial pass.

Article 2 — Jayanthi, Senthil Kumar & Gnanavadivel (2025): IHGM-SEPIC for DC Microgrids

The most immediate concern with the IHGM-SEPIC is its component count: thirteen elements (two MOSFETs, four inductors, four diodes, three capacitors) to deliver 24 V to 220 V conversion at 1 kW. For a single-phase variant the savings would be considerable. The interleaved arrangement does, however, buy something concrete. Against a conventional interleaved boost, the structure reduces maximum switch voltage by roughly half (the authors report 54.18 %) and lowers inductor inrush current by about a quarter (25.21 %). Switch voltage stress directly governs MOSFET selection and gate-drive losses; the inrush reduction matters at start-up and during PV irradiance transients.

Voltage gain follows from volt-second balance and resolves to $V_o/V_{in} = (1+d)/(1-d)$. The two phases conduct out of phase for one-third of the switching period each, producing four distinct operating modes. Simulated rated-load efficiency reaches 98.87 %, which the authors describe as 1.083 times the efficiency of an equivalent interleaved boost. The comparison is fair but narrow: an interleaved boost is not the natural competitor for a high-gain SEPIC. A more useful benchmark would be a coupled-inductor SEPIC at similar power level. The 24 kHz hardware prototype validates the analysis but does not push the design into a regime where the doubled component count would clearly pay off. As a DC microgrid converter for steady-irradiance operation, the topology works. Outside that regime the cost case is less clear.

2.2 Tuğrul Arslan (Student ID: 2004060)

Article 3 — Sumathy et al. (2024): HPSQB Quadratic-SEPIC for Water Treatment

Pulsed-electric-field water disinfection needs kilovolt pulses. A converter for that job has to push voltage gain into territory where conventional boost and basic SEPIC topologies stop making sense. Sumathy and co-authors propose the High-Pulse SEPIC–Quadratic Boost (HPSQB), which combines a quadratic boost stage, a SEPIC stage, and an extendable diode-capacitor multiplier cell. With one multiplier cell ($M = 1$), the device count is one MOSFET, three inductors, six capacitors, and six diodes, plus a pulse-forming switch.

Why extendable? Because each additional multiplier cell adds only two diodes and two capacitors. Switch voltage stress does not climb. That is the actual claim of the paper, and it lets the design reach a $17\times$ boost at $D = 0.61$, taking 300 V to 5.1 kV in simulation. The ideal gain is $(2+D)/(1-D)^2$.

The efficiency number is where things get interesting. At 500 W the analytical figure sits at 87 %, with a calculated loss of 67 W split across switches, diodes, capacitors and inductor resistances. That is honest but modest, especially for a pulse converter aimed at industrial use. The scaled-down 50 W prototype was tested with 12, 24, and 36 V inputs and run at 50 kHz, with the pulse-forming switch driven at 500 Hz. A real water sample was processed at 360 V and 5 kV in an outside laboratory.

The application angle is the rarer contribution. Most high-gain SEPIC papers stop at PV or microgrid framing. Linking the converter all the way to a microbial reduction test is unusual and gives the work additional weight. What is missing is a measured efficiency from the 50 W hardware. The theoretical 87 % needs experimental confirmation before the design can be called industrially ready.

Article 4 — Chandra & Gaur (2023): Switched Inductor-Capacitor SEPIC for Solar Applications

A switched inductor-capacitor cell mounted onto a classical SEPIC: that is the basis of the High Gain SEPIC (HGS) presented by Chandra and Gaur. No coupled inductor. No transformer. The motivation is direct. Leakage inductance creates voltage overshoot at turn-off, drives switch derating, and burns power as conduction loss. Removing it simplifies the magnetic design and the EMI shielding budget.

The full circuit needs one main switch, three inductors (L_1 , L_2 , L_M), four capacitors (C_1 , C_2 , C_S , C_o), and four diodes (D_1 , D_2 , D_S , D_o). In CCM the voltage gain is $2(1+D)/(1-D)$. At $D = 0.3$ the gain is 3.71; pushing D to 0.7 lifts it to 11.33. Analytical efficiency at 100 Ω load, 30 kHz switching, $D = 0.32$, and $V_{in} = 160$ V is 89.83 %, against 82.47 % for a classical SEPIC under matched conditions. A clear improvement, but not exceptional in this group.

Validation is done both ways: a laboratory prototype at a fixed 60 % pulse and a hardware-in-the-loop test with an OP4510 real-time simulator running P&O MPPT on a PV array. The HIL setup is a strong methodological choice for a PV-interface paper. The lab prototype, though, runs at fixed duty rather than under closed-loop tracking, and this is where the article underdelivers. The authors' bench conditions (room temperature, fixed load) do not capture industrial PV stress: temperature drift, partial shading, irradiance steps. Within its stated scope the paper makes its case. The $11\times$ gain without a coupled inductor is the practical result worth carrying forward.

2.3 Deniz Bayrak (Student ID: 2103588)

Article 5 — Abdi, Shokrani, Rostami & Babaei (2025): Coupled-Inductor SEPIC for Solar DC Microgrids

A $26\times$ voltage gain at $d = 0.5$ and turns ratio $n = 1$, derived from $V_o/V_{in} = (3+d+2n(2-d))/(1-d)^2$, places the Abdi–Shokrani–Rostami–Babaei converter at the top of the surveyed group on gain alone. The architecture combines a modified SEPIC body with a voltage doubler rectifier cell and a two-winding coupled inductor. According to the comparative table inside the paper, the device count totals sixteen: two MOSFETs, six diodes, six capacitors, one inductor, and one coupled inductor.

Several side-features carry weight. The MOSFET turn-on and turn-off transitions are synchronised by inherent circuit operation, so a complex auxiliary controller is unnecessary. Common ground between input and output is preserved, which improves EMI performance for PV interfaces. A pole-placement design handles the closed-loop dynamics. Efficiency at rated power is 95.24 % per the comparative table, with reported figures consistently above 95 % across operating points. The hardware prototype runs 25 V to 400 V at 200 W with 40 kHz switching. Experimental waveforms confirm step responses for 25 V $\rightarrow$ 15 V $\rightarrow$ 30 V at the input and 800 $\Omega \leftrightarrow$ 400 Ω load swings.

The peak gain figure relies on the coupled inductor. Removing the magnetic coupling drops the gain back toward the SEPIC-with-voltage-doubler baseline. The cost is the magnetic design itself: leakage management, EMI shielding, and saturation behaviour over the operating temperature range. None of these are addressed in measurement detail, only in nominal operation. The 16-component count is also non-negligible from a cost standpoint. The control derivation, by contrast, is unusually clean for a topology in this class. Per-device gain efficiency, computed in the discussion section, places this design at the top of the surveyed set.

Article 6 — Taheri, Baghrmian & Pourseyedi (2023): Three-Port High-Step-Up SEPIC

Three ports in one converter raises an immediate question about efficiency benchmarking, and the Taheri–Baghrmian–Pourseyedi article leaves it largely unanswered. The paper emphasises reduced switching and conduction losses qualitatively through low-stress switch operation. No single bottom-line efficiency number appears. For a topology meant to integrate a renewable source, a battery storage element, and a load, the absence of a quantitative efficiency benchmark weakens the case for adoption.

What the paper does deliver is structural. The three-port architecture supports two input ports (one bidirectional for battery, one for the renewable source) and one output port in a single converter, with three MOSFET switches, five diodes, one inductor, one coupled inductor, and four capacitors. The voltage lift cell (VLC) and the coupled inductor act as boost-augmenting structures. The renewable-port input current is continuous with low ripple.

Switch count is the architectural achievement. The active-switch total is the lowest among comparable multiport topologies in the reference set. Switch S1 sits behind a clamp circuit and sees low voltage stress; S2 and S3 benefit from the low-voltage character of the boosting cell, opening the door to lower- $R_{DS(on)}$ devices and reduced conduction loss. Two operating modes are considered. In SISO mode $V_o = (1+kN+d_{S1})/(1-d_{S1}) \times V_{in}$, where N is the turns ratio and k the coupling coefficient. The DISO expression extends to $V_o = (1+kN+d_{S1})/(1-d_{S1}) \times (V_{in1}+d_{S2} \cdot V_{in2})$, capturing simultaneous power from both inputs. A 100 W prototype at 22 V input, 600 Ω load, and 50 kHz switching validates the steady-state and transient analysis.

2.4 Defne Ceylan (Student ID: 2200666)

Article 7 — Mandal, Prabhakaran, Arun Dominic & Parameswaran (2025): Bidirectional ASL-SEPIC

Field experience with bidirectional power converters in EVs and microgrids tends to be unforgiving. Synchronous rectifier dead-time. Gate-driver losses at light load. Heatsink behaviour when the SiC MOSFETs run at 40 kHz for hours. The Mandal et al. paper takes these conditions seriously with a 200 W prototype built from UF3C065080K3S SiC devices and high-current drivers.

The topology is a non-isolated bidirectional converter. An Active Switched Inductor (ASL) sits at the input stage of a modified SEPIC core. Three switches, three inductors, three capacitors, no diodes. S3 acts as a synchronous rectifier instead. Replacing S3 with a passive diode collapses the structure to a unidirectional version. Boost-direction gain in CCM is $M = 2d/(1-d)$. Switch stress on S1 and S2 reads as $(1+M)/(2M) \times V_{DC-2}$, falling as gain rises.

Three numbers tell the efficiency story. Maximum theoretical: 96.8 %. Peak measured on the prototype: 95.4 %. Low end of the measured range: 85 %. That ten-percentage-point spread is the realistic operating envelope, and it lines up with what synchronous-rectifier converters typically show: peak efficiency at the optimal operating point, drop at light load where dead-time and gate-charge losses dominate.

For a vehicle-to-grid or grid-to-vehicle interface, bidirectional buck-boost capability with only three switches and no diodes is a real simplification. Component count stays low. Heat distribution is manageable because the ASL splits inductor currents between L1 and L2 nearly evenly across a wide range. The 85 % bottom-end efficiency suggests that low-load operating regions would benefit from a soft-switching modification before the design is ready for vehicle-grade duty cycles. As reported the converter works under bench conditions; the leap to transportation-grade reliability still needs evidence on thermal cycling and EMC compliance.

Article 8 — Jayanthi, Senthil Kumar, Gnanavadivel et al. (2024): Switched Inductor-Based HGSC

Compared against a classical SEPIC at the same duty cycle, the High Gain SEPIC Converter (HGSC) by Jayanthi et al. delivers 50 % more voltage gain and 50 % less voltage drop across the switches. The mechanism is the Switched Inductor Voltage Boosting Cell (SIVBC) inserted into the SEPIC body. The full device list is austere: two MOSFETs, three inductors, two capacitors, one diode.

That single diode is the practical highlight. Most high-gain SEPIC variants in this surveyed group use four to six diodes, which adds reverse-recovery losses and creates layout headaches at higher switching frequencies. Here the device count stays at eight. A 1 kW prototype delivers 380 V from a 48 V input. Switching frequency is 21 kHz: low for a modern design, deliberately chosen to keep gate-drive losses manageable. A PIC microcontroller generates the gating signals at an 80 % duty cycle.

In CCM the gain is $2D/(1-D)$. At $D = 0.85$ the gain reaches 11.33, which the authors quote as 35.58 % higher than classical SEPIC at the same duty. Simulated efficiency from the comparative table is 98.63 %; the abstract specifies a maximum measured efficiency of 97.88 %. A synchronous-reference-frame current controller manages power flow into a grid-tied inverter, with incremental conductance MPPT tracking the PV peak power.

The 85 % duty cycle is the part that demands caution in actual deployment. It means the switch is on for most of the period, leaving little margin for transient over-currents or dynamic load steps. PIC-based gating at 21 kHz is robust but constrains controller speed. As a DC microgrid converter under steady irradiance the design holds up. Under cloud-cover transients the duty headroom is the variable to watch. From a parts-cost and footprint perspective, the single-diode arrangement is the most compact in this study.

3. Comparison Table

The table below summarises the eight converters along sixteen columns. The "GCR" column reports a derived metric introduced for this review: Gain-to-Component Ratio, computed as the voltage gain at the operating point cited in each article divided by the total device count (switches + inductors + capacitors + diodes, with a coupled inductor counted as one device). The "Validation" column compresses how each result was generated: Sim = simulation, Exp = hardware prototype, HIL = hardware-in-the-loop. Component counts and reported figures come directly from the articles. Abbreviations: CI = coupled inductor; N/R = not reported; PEF = pulsed electric field; BBB = buck-boost bidirectional.

Ref.	Topology	Sw	L	C	D	Voltage Gain	Max Gain (D)	GCR	Sim η (%)	Exp η (%)	V _{in} / V _{out}	Power	f _{sw}	Validation	Target Application
[1] Baladhandapani 2025	Interleaved SEPIC-Cuk	2	4	3	2	$2/(1-D)$	10.2 (D=0.585)	0.93	92.0	N/R	65 V / 350 V	~765 W	10 kHz	Sim + Exp	PV-fed PMSM EV
[2] Jayanthi 2025	IHGM-SEPIC	2	4	3	4	$(1+d)/(1-d)$	9.17 (rated)	0.71	98.87	N/R	24 V / 220 V	1 kW	24 kHz	Sim + Exp	DC microgrid PV
[3] Sumathy 2024	HPSQB (M=1)	1	3	6	6	$(2+D)/(1-D)^2$	17 (D=0.61)	1.06	87 (analyt.)	N/R	300 V/5 kV (sim); 12–36 V (proto)	500 W / 50 W	50 kHz	Sim + Exp	Water treatment (PEF)
[4] Chandra 2023	HGS SI-C SEPIC	1	3	4	4	$2(1+D)/(1-D)$	11.33 (D=0.7)	0.94	89.83	N/R	160 V / variable	variable	20–30 kHz	Sim + HIL + Exp	Solar PV
[5] Abdi 2025	CI-SEPIC + volt. doubler	2	1+CI	6	6	$(3+d+2n(2-d))/(1-d)^2$	26 (d=0.5, n=1)	1.63	95.24	>95 (multi-pt)	25 V / 400 V	200 W	40 kHz	Sim + Exp	Solar DC microgrid
[6] Taheri 2023	Three-port SEPIC + VLC + CI	3	1+CI	4	5	$(1+kN+d_{S1})/(1-d_{S1})$	9 (SISO repr.)	0.64	N/R	N/R	22 V / variable	100 W	50 kHz	Sim + Exp	Multi-input renewable
[7] Mandal 2025	Bidirectional ASL-SEPIC	3	3	3	0 (sync.)	$2d/(1-d)$	8 (d=0.8, BBB)	0.89	96.8 (theor.)	85–95.4	bidirectional	200 W	40 kHz	Sim + Exp	EV, microgrid, medical
[8] Jayanthi 2024	HGSC + SIVBC	2	3	2	1	$2D/(1-D)$	11.33 (D=0.85)	1.42	98.63	97.88 (max)	48 V / 380 V	1 kW	21 kHz	Sim + Exp	DC microgrid solar PV

4. Discussion and Observations

Two patterns surfaced once the numbers were laid out side by side. The converter with the highest reported simulation efficiency in this group (Jayanthi 2025 IHGM at 98.87 %) is also the one with the highest component count among the non-coupled designs (thirteen elements). High efficiency was bought through interleaving, not topological simplicity. The converter with the highest voltage gain (Abdi 2025 at $26\times$) shares the same device count of sixteen as the Sumathy water-treatment converter but reaches more than 50 % higher gain. Magnetic coupling, not device count, drives that outcome.

To make this kind of comparison concrete, the Gain-to-Component Ratio (GCR) introduced in the table places the eight designs in the following order, highest first: Abdi 2025 at 1.63, Jayanthi 2024 HGSC at 1.42, Sumathy 2024 at 1.06, Chandra 2023 at 0.94, Baladhandapani 2025 at 0.93, Mandal 2025 at 0.89, Jayanthi 2025 IHGM at 0.71, and Taheri 2023 at 0.64 in SISO mode. Two clusters are visible. Coupled-inductor designs with multiplier cells (Abdi) sit at the top because the turns ratio acts as a separate gain lever beyond what the duty cycle alone provides. Switched-inductor designs with minimal device counts (Jayanthi 2024 HGSC) come second for a different reason: not because their gain is highest, but because their denominator is smallest. The lower half of the GCR ranking is dominated by topologies that pay an extra component price for application features rather than gain, namely interleaving (Jayanthi IHGM) and multi-port architecture (Taheri).

A caveat. The GCR metric is not entirely fair. Voltage gain is duty-dependent, so the comparison picks one operating point for each design. Taheri's three-port topology in particular looks penalised under this normalisation because its multi-input flexibility is what justifies its component count, not its raw boost ratio. A more honest reading is that GCR works as a comparison axis for the four single-source non-coupled designs in the set, where the operating conditions are similar enough for direct comparison.

Efficiency observations split along application lines. The DC microgrid converters (Jayanthi 2024, Jayanthi 2025, Chandra, Abdi) sit between 89.83 % and 98.87 % in simulation. The EV-targeted converters (Baladhandapani, Mandal) fall to 92.02 % and the 85–95.4 % measured band respectively, reflecting the harder operating envelope that vehicle applications impose. Sumathy's water-treatment converter at 87 % analytical efficiency is the outlier on the low side, but the application demands kilovolt outputs and pulsed operation puts converters under fundamentally different stress conditions.

The validation column in the table is worth a separate comment. Three of the eight articles rely heavily on simulation, with prototypes used mainly to confirm waveform shapes (Sumathy via nL5, Baladhandapani via MATLAB Simscape, Jayanthi 2025 in part). Four report both simulation and full hardware characterisation (Abdi, Mandal, Jayanthi 2024, Taheri). Chandra adds a hardware-in-the-loop layer with an OP4510 real-time simulator, which is methodologically the strongest validation in the set. For research applications that target deployment, HIL captures closed-loop behaviour more faithfully than fixed-duty bench testing. The reader has to weigh validation depth alongside the quoted efficiency numbers, because not all 95 % efficiencies are equally trustworthy.

A final observation about operating conditions. Most of the experimental data in these eight articles was collected at room temperature, under steady load, with controlled input. Industrial PV interfaces add temperature drift, partial shading, irradiance steps, and inverter-side disturbances. None of the surveyed papers report measurements under those conditions. The reported efficiencies are therefore upper bounds on field performance, not predictions of it.

The eight converters do not converge on a single best topology. They divide the design space. Coupled-inductor variants reach the highest gains but inherit magnetic design complexity. Switched-inductor variants stay simpler but cap out near $11\text{--}12\times$ gain. Multiport designs add system-level flexibility at the cost of per-port efficiency reporting. The match between topology and application is the variable to optimise, not gain or efficiency in isolation.

5. Conclusion

The eight high-gain SEPIC variants reviewed here span the 2023–2025 window and divide along three lines: coupled-inductor designs (Abdi, Taheri) that reach the highest gains but carry magnetic design complexity, switched-inductor designs (Jayanthi 2024 HGSC, Chandra, Jayanthi 2025 IHGM) that stay simpler but cap below $12\times$ gain, and mixed structures (Sumathy quadratic-SEPIC, Mandal ASL-SEPIC, Baladhandapani SEPIC-Cuk) that target specific applications. Reported gains range from $9\times$ to $26\times$; reported efficiencies range from 87 % analytical to 98.87 % simulated. The Gain-to-Component Ratio metric introduced in the discussion places Abdi's coupled-inductor topology at the top (1.63) and Jayanthi 2024's HGSC second (1.42) by a substantial margin, suggesting that magnetic coupling and minimal device count both serve gain efficiency, in different ways. No single converter dominates across applications. Topology selection should follow the application profile.

6. References

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